

Figure 4: Docker based on LXC

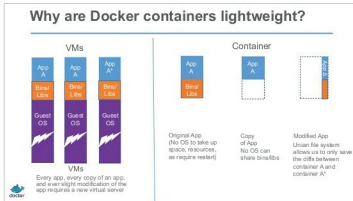


Figure 5: Docker containers are lightweight

faster while remaining secure. It is associated with service provider tools, dev tools, official repositories, orchestration tools, systems integration tools, service discovery tools, Big Data, security tools, monitoring and logging tools, configuration management tools such as those used for continuous integration, etc.

Continuous integration (CI) is another big area for Docker. Traditionally, CI services have used VMs to create the isolation you need to fully test a software app. Docker's containers let you do this without using a lot of resources, which means your CI and your build pipeline can move more quickly.

Continuous integration and continuous deployment (CD) have become one of the most common use cases of Docker

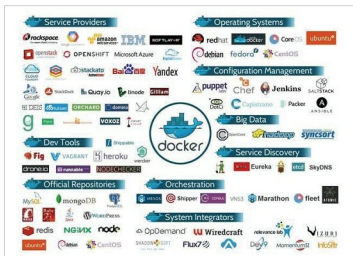


Figure 6: Docker associated tools

early adopters. CI/CD merges development with testing, allowing developers to build code collaboratively, submit it to the master branch and check for issues. This allows developers to not only build their code, but also test it in any environment type and as often as possible to catch bugs early in the applications development life cycle. Since Docker can integrate with tools like Jenkins and GitHub, developers can submit code to GitHub, test it and automatically trigger a build using Jenkins. Then, once the image is complete, it can be added to Docker registries. This streamlines the process and saves time on build and set-up processes, all while allowing developers to run tests in parallel and automate them so that they can continue to work on other projects while tests are being run. Since Docker works on the cloud or virtual environment and supports both Linux and Windows, enterprises no longer have to deal with inconsistencies between different environments – which is perhaps one of the most widely known benefits of the Docker CaaS (Containers as a Service) platform.

Drone.io is a Docker-specific CI service, but all the big CI players have Docker integration anyway, including Jenkins, Puppet, Chef, Saltstack, Packer, Ansible, etc; so it will be easy to find and incorporate Docker into your process.

Adoption of Docker

Docker is probably the most talked about infrastructure technology in the past few years. A study by Datadog, covering around 10,000 companies and 185 million containers in real-world use, has resulted in the largest and most accurate

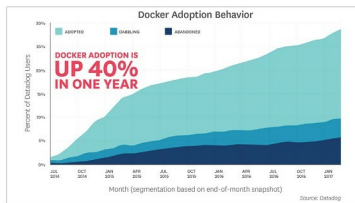


Figure 7: Docker adoption has increased 40 per cent in one year

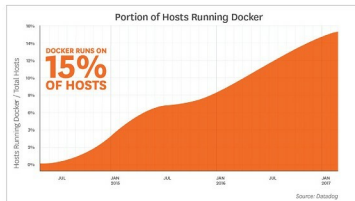


Figure 8: Docker now runs on 15 percent of the hosts